

INTRODUCTION

2.1 Background

Radiation monitoring is an essential aspect of nuclear safety, environmental protection, healthcare, industrial applications, and emergency response. Traditional radiation monitoring techniques generally provide radiation measurements at fixed locations or through handheld instruments, making it difficult to visualize radiation distribution over a large geographical area in real time. These conventional methods also lack the capability to accurately identify the location of radioactive sources and provide limited support for rapid decision-making during radiological emergencies.

With the rapid advancement of embedded systems, Global Positioning System (GPS) technology, Internet of Things (IoT), and geospatial visualization techniques, intelligent radiation mapping systems have become increasingly important. Integrating radiation detectors with GPS modules and edge computing platforms such as Raspberry Pi enables the collection of radiation measurements together with precise geographical coordinates. These datasets can then be visualized as interactive heatmaps, providing valuable insights into radiation distribution across a monitored area.

The **RADMAP (RADiation MAPing)** project was developed to address these challenges by creating a portable GPS-enabled radiation mapping prototype capable of acquiring real-time radiation data, environmental parameters, and geographical information. The system integrates hardware sensors with software visualization tools to provide accurate, scalable, and intelligent radiation monitoring for future nuclear safety applications.

2.2 Project Motivation

The primary motivation behind this project was to overcome the limitations of conventional **2D radiation heatmaps**. Although heatmaps effectively represent radiation intensity across different locations, they do not provide information regarding the exact physical location of the radioactive source. Identifying the origin of radiation is one of the major challenges in radiation monitoring systems.

To address this research problem, the project focused on extending conventional 2D radiation mapping into a **3D Digital Twin environment** capable of supporting future radioactive source localization. The motivation was further strengthened by the need to integrate live GPS coordinates, environmental sensing, camera feedback, and radiation measurements into a unified visualization platform.

The internship also provided an opportunity to gain practical experience in embedded system development, Raspberry Pi programming, GPS telemetry, radiation instrumentation, data visualization, backend development, and research-based engineering. The project combines software development with real-world nuclear technology applications, contributing toward intelligent radiation monitoring and environmental surveillance.

2.3 Project Scope

The scope of the RADMAP project includes the complete development of a GPS-enabled radiation mapping system using Raspberry Pi-5 and multiple integrated hardware components. The project involved developing structured radiation datasets, generating interactive 2D radiation heatmaps, integrating the GlobalSat BU-353N GPS Receiver for real-time latitude, longitude, temperature, humidity, and timestamp acquisition, and synchronizing this information with Raspberry Pi-5.

The project further includes the integration of a **Gamma Spectrometer (CNSPEC MCA)** for radiation detection and validation using standard radioactive sources such as **Cobalt-60 (Co-60)** and **Cesium-137 (Cs-137)**. Based on the acquired radiation measurements, dynamic heatmaps displaying radiation counts, peak intensity, and spatial distribution were successfully generated.

Another major scope of this project is the research and development of a **3D Digital Twin** for future radioactive source localization. The developed system integrates live camera feedback, GPS telemetry, and radiation data to provide real-time three-dimensional visualization. In addition, a backend architecture was developed to synchronize sensor communication, environmental data, radiation processing, and visualization modules. A night-vision-inspired

color grading algorithm was also implemented to represent varying radiation intensity levels on live camera feeds, improving situational awareness during radiation monitoring.

The outcomes of this project establish a strong foundation for future enhancements, including AI-assisted source localization, LiDAR-SLAM integration, autonomous robotic navigation, drone-based radiation mapping, and cloud-based radiation monitoring systems for nuclear safety applications.

2.4 My Role in the Project

During the internship, I actively contributed to both the research and development phases of the RADMAP project. My work included developing radiation datasets and publishing them on Kaggle, designing interactive 2D radiation heatmaps, integrating the GlobalSat BU-353N GPS receiver with Raspberry Pi-5 for real-time telemetry acquisition, and processing environmental parameters such as latitude, longitude, temperature, and humidity.

I also worked on integrating and validating the Gamma Spectrometer using Co-60 and Cs-137 radioactive sources, developed the backend architecture for data acquisition and visualization, researched and implemented the concept of a 3D Digital Twin for radiation source localization, designed a real-time color grading system inspired by night-vision imaging, and created a 3D printable prototype model of the RADMAP system. These contributions significantly enhanced the functionality of the prototype and supported its approval for future funding and advanced research.